FIAP

GRADUAÇÃO 2018

Dynamic Systems & Control

Docente: Christiam S. M. Alvarado

Temário

- Introdução
- Definição da Função de Transferência (FT)
- Diagrama de Blocos
- Álgebra de Blocos

Introdução

- Diversas plantas industriais são bem representadas por funções de transferência (modelos matemáticos).
- Na teoria de controle, as funções de transferência são comumente utilizadas para caracterizar as relações de entrada-saída de componentes ou sistemas que podem ser descritos por equações diferenciais.



Definição - Função de Transferência



Definição - Função de Transferência

- Considere um sistema linear, invariante no tempo, a parâmetros concentrados descrito pela seguinte equação diferencial:

$$y^{(n)} + a_1 y^{(n-1)} + \ldots + a_{n-1}\dot{y} + a_n y = b_1 u^{(n-1)} + b_2 u^{(n-2)} + \ldots + b_{n-1}\dot{u} + b_n u$$

- Aplicando a transformada de Laplace em ambos os lados da equação acima, com condições iniciais nulas:

$$\left(s^n + a_1 s^{n-1} + \ldots + a_{n-1} s + a_n\right) Y(s) = \left(b_1 s^{n-1} + b_2 s^{n-2} + \ldots + b_{n-1} s + b_n\right) U(s)$$

$$FT = G(s) = \frac{Y(s)}{U(s)} = \frac{\left(b_1 s^{n-1} + b_2 s^{n-2} + \ldots + b_{n-1} s + b_n\right)}{\left(s^n + a_1 s^{n-1} + \ldots + a_{n-1} s + a_n\right)}$$

Definição - Função de Transferência

- Também, a Função de Transferência pode ser escrita como:

$$G(s) = \frac{K(s - z_1)(s - z_2)\dots(s - z_{n-1})}{(s - p_1)(s - p_2)\dots(s - p_n)} = K\frac{N(s)}{D(s)}$$

Onde:

$z_1, z_2, \dots, z_{n-1}$ são os zeros do sistema $G(s) = 0$

$p_1, p_2, \dots, p_n$ são os pólos do sistema $G(s) \to \infty$

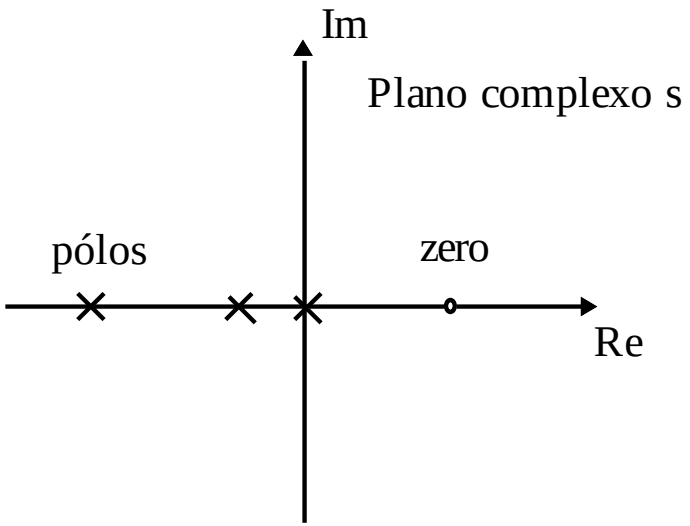
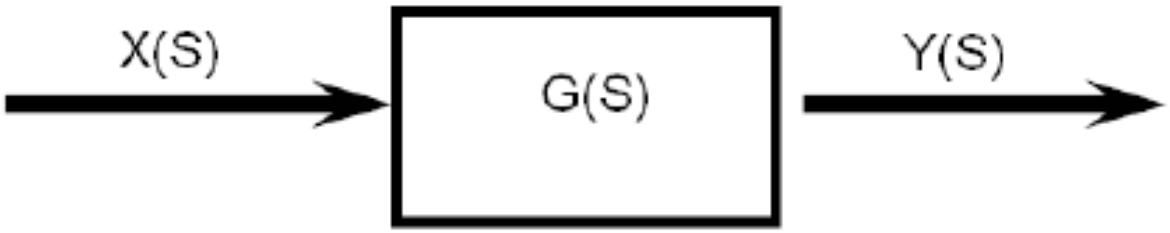


Diagrama de Blocos

Fornece uma visão gráfica global do sistema indicando realisticamente a finalidade dos componentes dentro do sistema, e como ocorre o fluxo de sinais entre os blocos.

$$G(s) = Y(s) / X(s) \Rightarrow Y(s) = G(s).X(s)$$



X(S) – Transformada de Laplace da entrada.

Y(S) – Transformada de Laplace da saída.

Diagrama de Blocos - Elementos

- Bloco funcional: Apresenta no seu interior a função de transferência.
- Ponto de soma: indica uma operação de adição ou subtração.
- Ponto de junção: ponto a partir do qual o sinal proveniente de um bloco vai para outros blocos ou pontos.

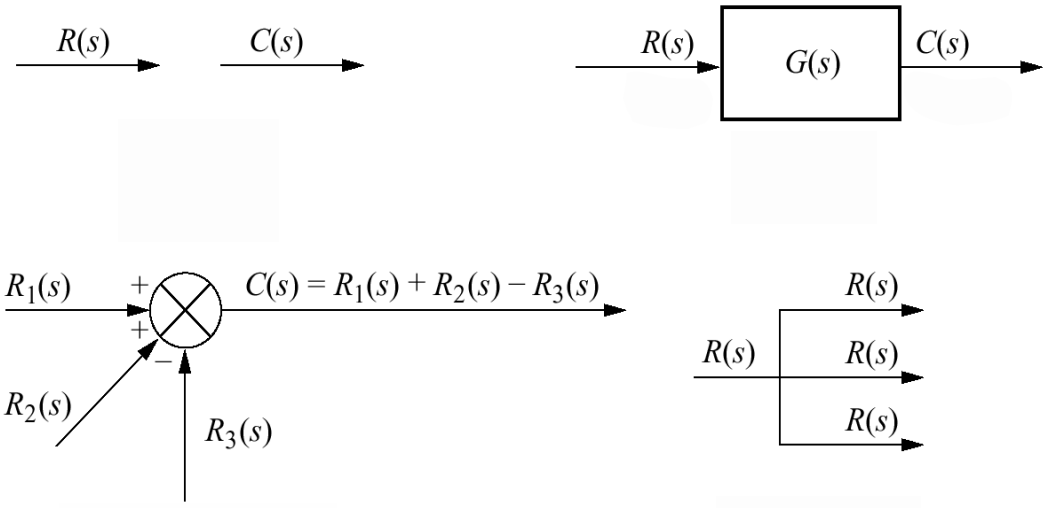


Diagrama de Blocos – Equivalência

Função de transferência e diagrama de blocos equivalentes

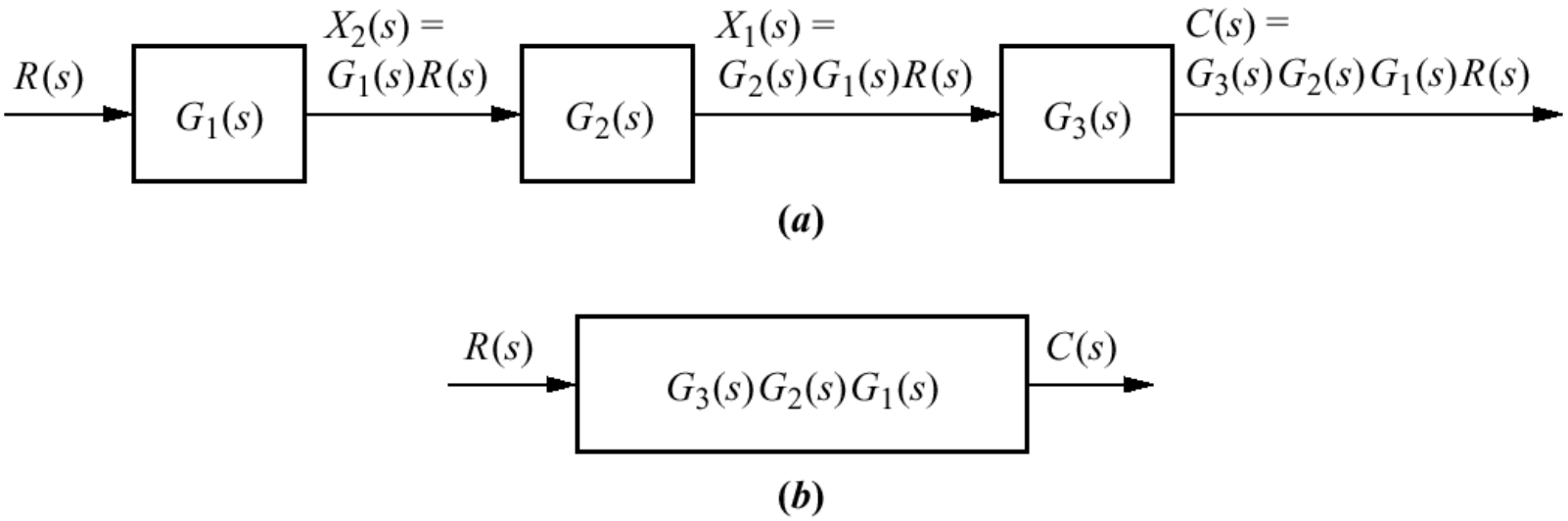
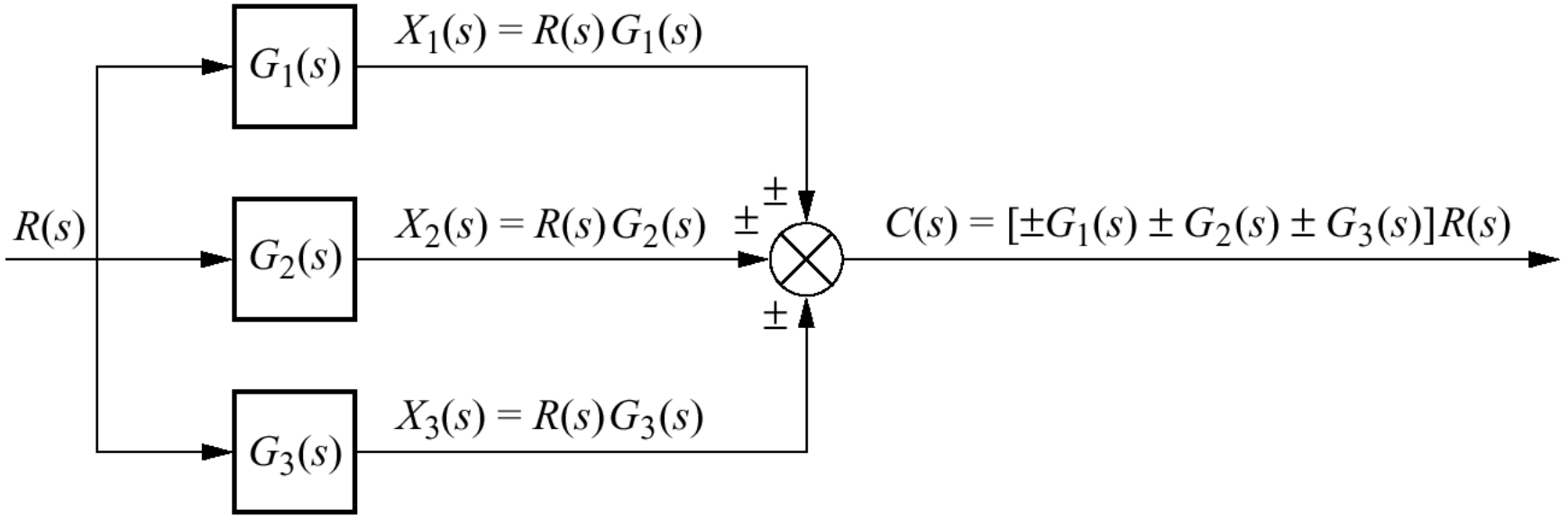
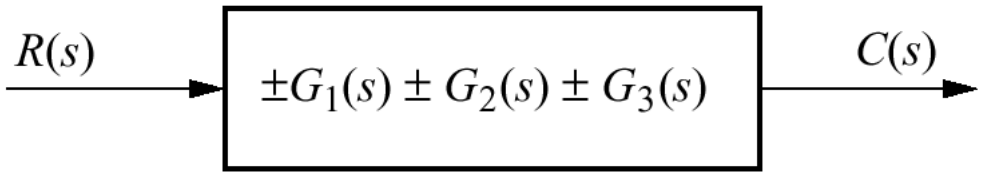


Diagrama de Blocos - Equivalências

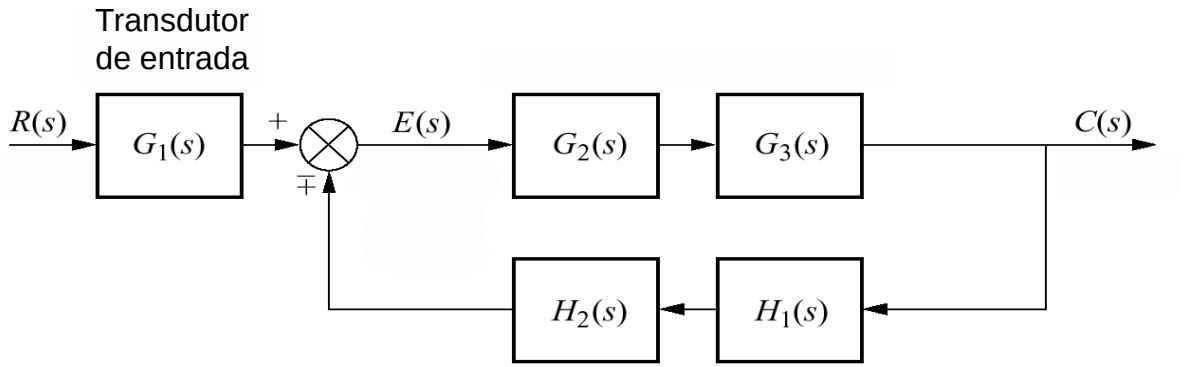


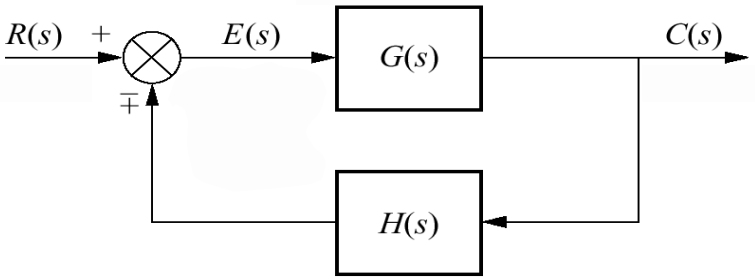
(*a*)



(*b*)

Diagrama de Blocos - Equivalências





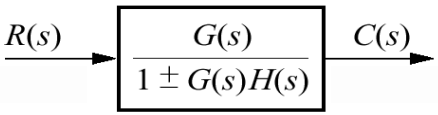
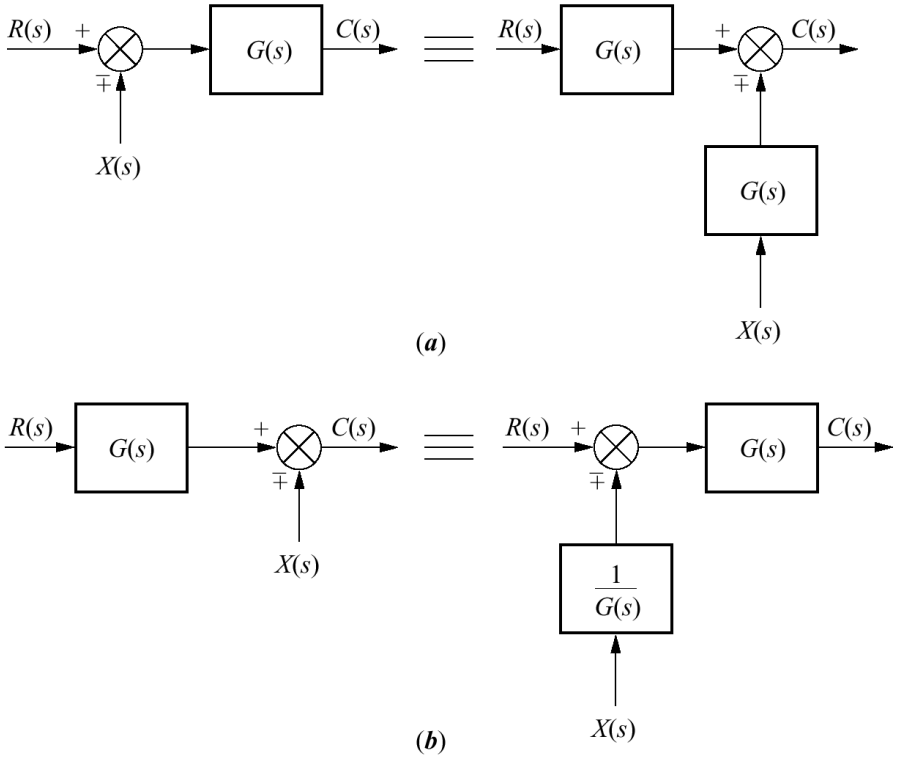
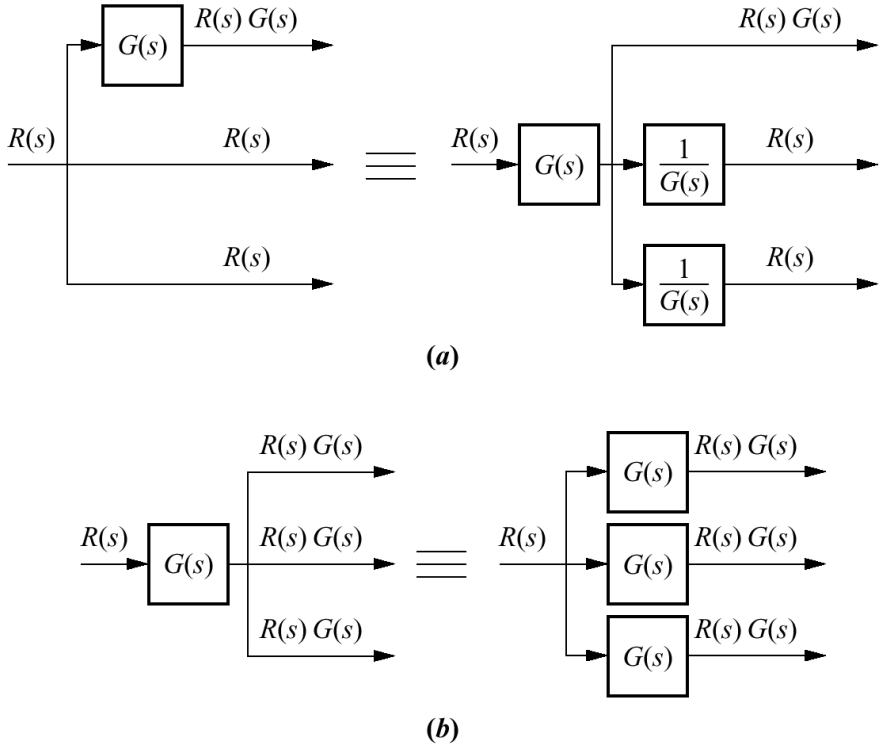


Diagrama de Blocos - Reduções

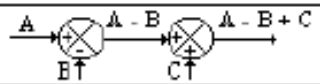
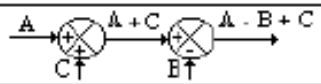
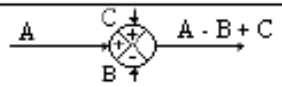
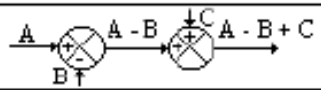
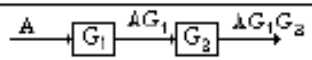
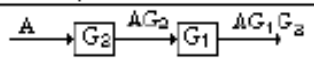
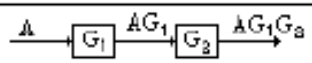
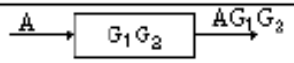
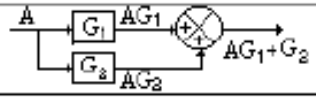
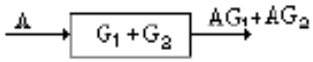
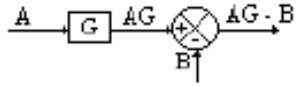
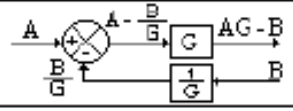
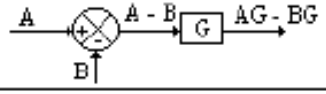
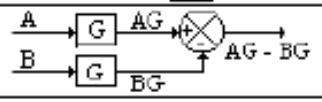
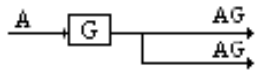
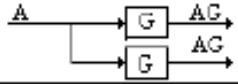
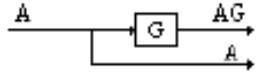
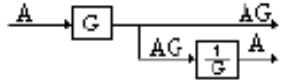
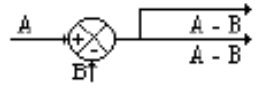
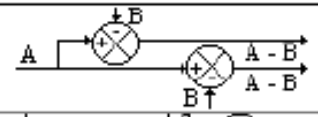
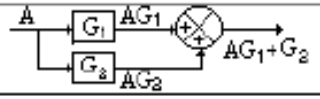
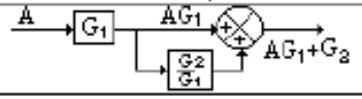
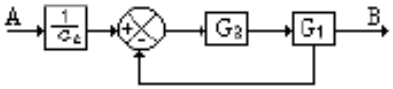
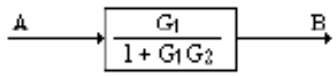


Álgebra de diagrama de blocos para junções de soma – formas equivalentes de deslocar um bloco



Álgebra de diagrama de blocos para junções de aquisição de sinais – formas equivalentes de deslocar um bloco

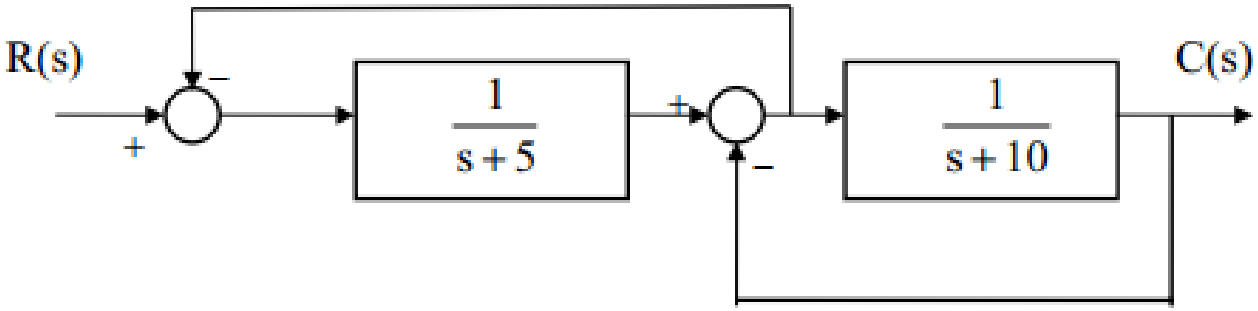
Diagrama de Blocos – Tabela de Equivalências

	DIAGRAMA BLOCO ORIGINAL	DIAGRAMA BLOCO EQUIVALENTE
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		

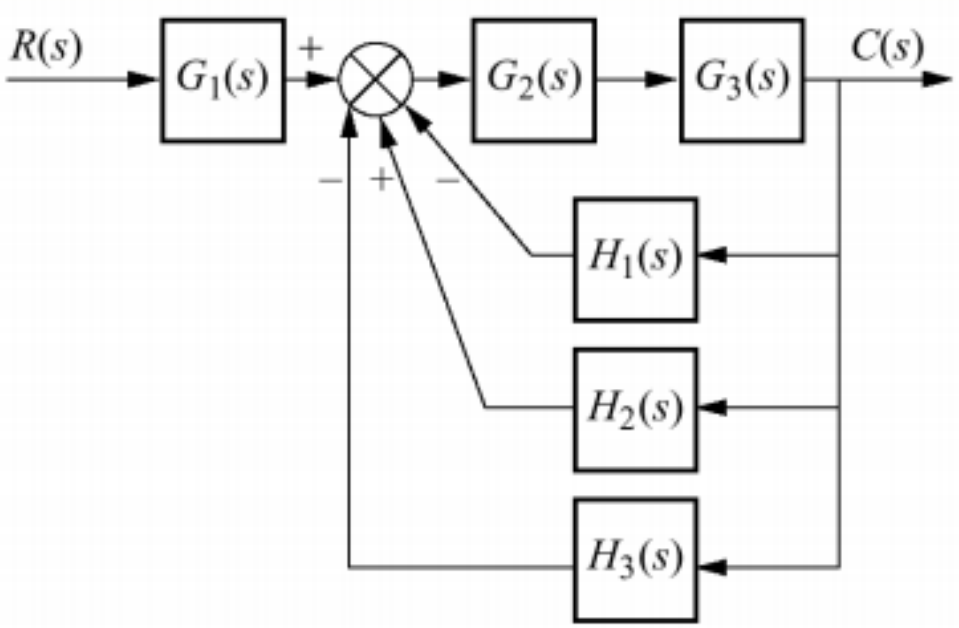
Exercícios

Determinar a FT *C(s)/R(s)*

1.

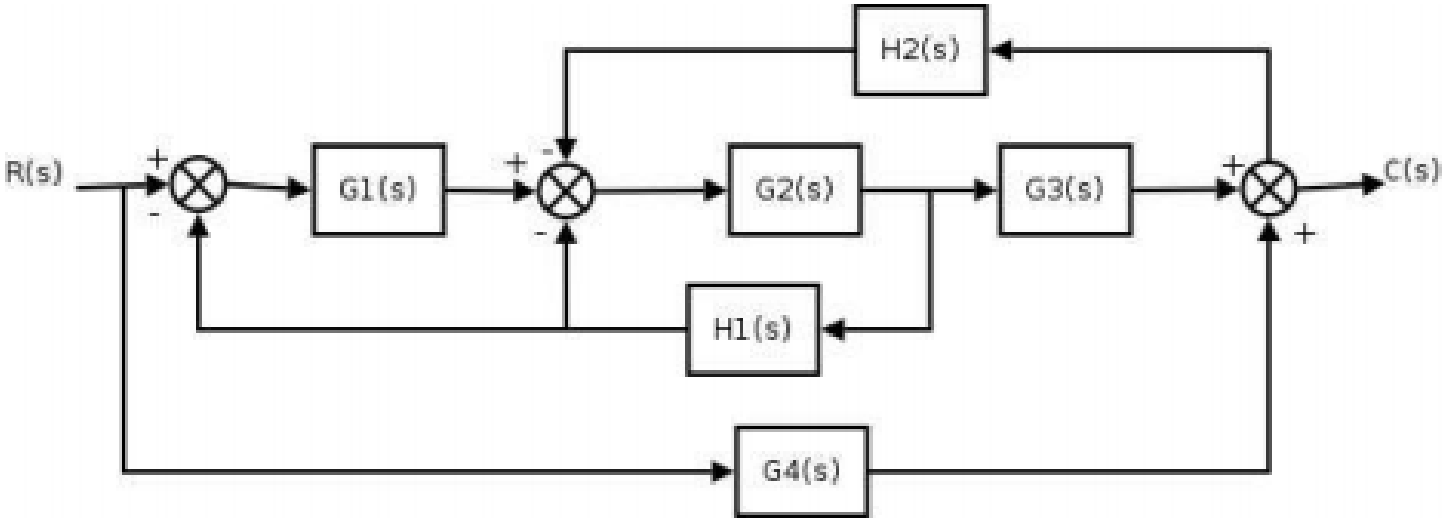


2.

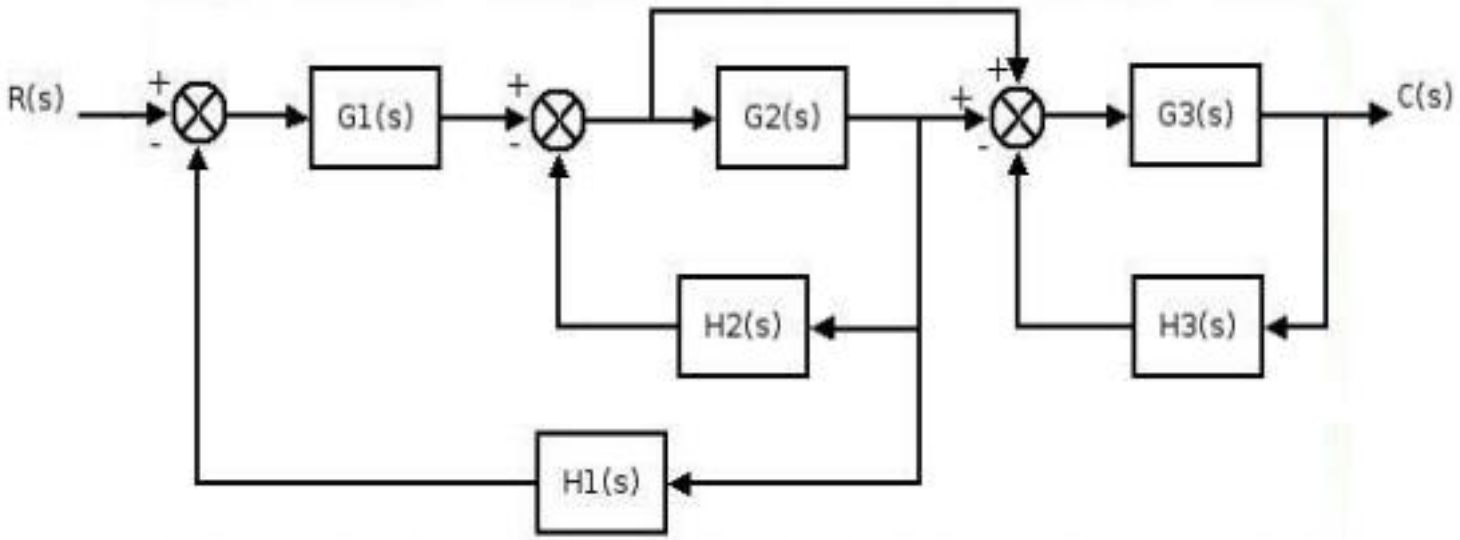


Exercícios

3.



4.



FIAP

GRADUAÇÃO
2018

Copyright © **2018** Prof.

Todos direitos reservados. Reprodução ou divulgação total ou parcial deste documento é expressamente proíbido sem o consentimento formal, por escrito, do Professor (autor).